

● EASY

● ALGEBRA

● ANSWER KEY

Linear functions

02

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**A**

A. $f(x) = \frac{1}{4}x + 5$

Choice A is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m is the slope of the graph of $y = f(x)$ in the xy -plane and $(0, b)$ is the y -intercept of the graph. It's given that for the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of $\frac{1}{4}$. Therefore, $m = \frac{1}{4}$. It's also given that the graph of $y = f(x)$ passes through the point $(0, 5)$. Therefore, the y -intercept of the graph is $(0, 5)$, and it follows that $b = 5$. Substituting $\frac{1}{4}$ for m and 5 for b in the equation $f(x) = mx + b$ yields $f(x) = \frac{1}{4}x + 5$.

Choice B is incorrect. This equation defines a function whose graph has a y -intercept of $(0, \frac{1}{5})$, not $(0, 5)$.

Choice C is incorrect. This equation defines a function whose graph has a y -intercept of $(0, -\frac{5}{4})$, not $(0, 5)$.

Choice D is incorrect. This equation defines a function whose graph has a y -intercept of $(0, -5)$, not $(0, 5)$.

Q2**C**

C. $c = 23t + 35$

Choice C is correct. It's given that this service contract requires a monthly cost of \$ 23. A monthly cost of \$ 23 for t months results in a cost of \$ $23t$. It's also given that this service contract requires a onetime activation cost of \$ 35. Adding the onetime activation cost to the monthly cost of the service contract for t months yields the total cost c , in dollars, of this service contract for t months. Therefore, this situation can be represented by the equation $c = 23t + 35$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**A****A. \$46.60**

Choice A is correct. It's given that Robert was charged an initial fee of \$20.00 to rent the truck plus an additional \$0.70 per mile driven. Let m represent the number of miles the truck was driven. Since the rental charge is \$0.70 per mile driven, $0.70m$ represents the amount Robert was charged for m miles driven. Let c equal the total amount, in dollars, Robert was charged to rent the truck. The total amount can be represented by the equation $c = 20.00 + 0.70m$. It's given that the truck was driven 38 miles, thus $m = 38$. Substituting 38 into the equation gives $c = 20.00 + 0.70(38)$. Multiplying $0.70(38)$ gives $c = 20.00 + 26.60$. Adding these values gives $c = 46.60$, so the total amount Robert was charged is \$46.60.

Choices B, C, and D are incorrect and may result from setting up the equation incorrectly or from making calculation errors.

Q4**B****B. 32**

Choice B is correct. Since Brittany's total take-home pay was \$1,576, the value 1,576 can be substituted for T in the given equation $T = 1,000 + 18h$ to give $1,576 = 1,000 + 18h$. Subtracting 1,000 from both sides of this equation gives $576 = 18h$. Dividing both sides of this equation by 18 gives $32 = h$. Therefore, Brittany was paid for 32 hours for her first week of work.

Choice A is incorrect. This is half the number of hours Brittany was paid for. Choice C is incorrect and may result from dividing 1,000 by 18. Choice D is incorrect and may result from dividing 1,576 by 18.

Q5**A**

A. The estimated height, in cm, of the liquid at the start of the project

Choice A is correct. It's given that the function $f(x) = 33 - 0.18x$ gives the estimated height, in centimeters (cm), of the liquid in the container x days after the start of the project. For a linear function in the form $f(x) = a + bx$, where a and b are constants, a represents the value of $f(0)$ and b represents the rate of change of the function. It follows that in the given function, 33 represents the value of $f(0)$. Therefore, the best interpretation of 33 in this context is the estimated height, in cm, of the liquid at the start of the project.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. The estimated change in the height, in cm, of the liquid each day is 0.18, not 33.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**A**

A. -34

Choice A is correct. It's given that the function g is defined by $g(x) = 4x - 6$. Substituting -7 for x into the given equation yields $g(-7) = 4(-7) - 6$, or $g(-7) = -34$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of x for which $g(x) = -7$, not the value of $g(-7)$.

Q7**20**

The correct answer is 20. For the graph shown, the x -axis represents the time since the initial deposit, in months, and the y -axis represents the bank account balance, in dollars. The amount of the initial deposit is estimated by the y -coordinate of the point on the graph that represents 0 months since the initial deposit. Therefore, the amount of the initial deposit is estimated by the corresponding y -value for the point when $x = 0$. When $x = 0$, it is estimated that $y = 20$. Thus, the amount of the initial deposit estimated by the graph, to the nearest whole dollar, is 20.

Q8**B**

B. $f(x) = \frac{3}{4}x + 3$

Choice B is correct. The equation for the function f in the xy -plane can be represented by $f(x) = mx + b$, where m is the slope and b is the y -coordinate of the y -intercept. Since it's given that the line has a slope of $\frac{3}{4}$, it follows that $m = \frac{3}{4}$ in $f(x) = mx + b$, which yields $y = \frac{3}{4}x + b$. It's given that $f(0) = 3$. This implies that the graph of the function f in the xy -plane passes through the point $(0, 3)$. Thus, the y -coordinate of the y -intercept of the graph is 3, so $b = 3$ in $f(x) = \frac{3}{4}x + b$, which yields $f(x) = \frac{3}{4}x + 3$. Therefore, the equation $f(x) = \frac{3}{4}x + 3$ defines the function f .

Choice A is incorrect and may result from a sign error for the y -intercept. Choice C is incorrect and may result from using the denominator of the given slope as m in $f(x) = mx + b$, in addition to a sign error for the y -intercept. Choice D is incorrect and may result from using the denominator of the given slope as m in $f(x) = mx + b$.

Q9**D**

D. 0,8

Choice D is correct. The y -intercept of a graph is the point where the graph intersects the y -axis. The graph of function f shown intersects the y -axis at the point $(0, 8)$. Therefore, the y -intercept of the graph of f is $(0, 8)$.

Choice A is incorrect. This is the point where the x -axis, not the graph of f , intersects the y -axis.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q10**B**

B. 0, - 4

Choice B is correct. The y -intercept of a graph is the point where the graph intersects the y -axis. The graph of function f shown intersects the y -axis at the point $(0, -4)$. Therefore, the y -intercept of the graph is $(0, -4)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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